

Asmaa Abdul-Amin

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Summary

Previously held U.S. Secret Security Clearance

Software Engineer with 5+ years of experience developing web applications, backend systems, automation tools, and data-driven software through freelance consulting and the Johns Hopkins Applied Physics Laboratory. Experienced in Python, SQL, JavaScript, REST APIs, machine learning, and secure software engineering. Passionate about building scalable software for aerospace, defense, and artificial intelligence applications.

Experience

Freelance Software Engineer

Self-Employed – 2021 - Present

- Design and develop custom web applications and backend systems using **Python**, **JavaScript**, and **SQL**.
- Build **REST APIs**, automation tools, and data processing pipelines.
- Design relational databases and integrate third-party APIs.
- Deploy applications using **Linux**, **Docker**, and **cloud services**.
- Manage projects from client requirements through deployment and ongoing maintenance.

Johns Hopkins Applied Physics Laboratory

March 2023 – May 2026

Computer Science/Computer Engineering Intern – Pathways Program

The Johns Hopkins Applied Physics Laboratory – Space Exploration Sector (**SOF-2**) | May 2025 – May 2026

- Continuing to fortify and **optimize Space Sector web infrastructure**, boosting reliability by **~40%** and ensuring compliance across **15+** NASA mission platforms.
- Automated Python + SQL workflows**, cutting data processing time by **60%** and saving analysts **15+** hours weekly.
- Collaborated with cross-functional teams and presented findings to stakeholders, ensuring **data-driven solutions aligned with mission and business needs**

Software & Web Infrastructure Intern – ATLAS Program

The Johns Hopkins Applied Physics Lab – SES/SOF-2 | May 2024 – May 2025

- Contributed to an **automated ETL pipeline** for mission requirements/verifications database, strengthening traceability and system interoperability across mission platforms supporting 100+ engineers, demonstrating skills in data ingestion, data manipulation, and data orchestration for high-frequency environments.
- Continued to support the enhancement of APL space sector **web infrastructure modernization** efforts across multiple mission platforms, ensuring secure, scalable, and compliant backend operations.

Undergraduate Research Intern – CIRCUIT Program

Johns Hopkins APL – Research & Exploratory Development Department (R1N) | Mar 2023 – May 2024

- Engineered and **deployed multiple webpages for the NASA Dragonfly** team portal, streamlining access to critical mission resources and improving mission collaboration efficiency.
 - Developed ML model analyzing COVID-19 policies** vs. mortality, improving interpretability of public health data.
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Projects

[DIAMOND HANDS](#) – Group project identifying resilient, high-growth stocks using data-driven financial analysis.

[QuantumQuant](#) – A modular Python **trading platform** integrating automated analytics for real-time portfolio monitoring and trade execution.

[CryoET](#) – Trained PyTorch deep learning models for **3D protein detection**, improving recall in high-dimensional biomedical datasets.

[The Cost of Democracy](#) – Analysis of product **price fluctuations across election cycles** revealing trends in consumer costs, inflation, and market response.

Education

B.S. Data Science – University of Maryland Global Campus (Expected Dec 2028)

Focus: Data Analysis, Machine Learning, and Algorithmic Systems, Python Programming

A.A. Computer Science – Montgomery College (Expected Dec 2026)

Certifications:

[The Johns Hopkins University Applied Physics Laboratory – Python Programming II \(2025\)](#)

[Univ. of Chicago – Quantum Computing Systems Design I \(2024\)](#)

[IBM – Machine Learning with Python \(2023\)](#)

[Harvard – CS50 Intro to Computer Science \(2022\)](#)

Skills

- Libraries & Tools:** Atlassian, Bitbucket, Docker, Git, Linux, Jupyter, HuggingFace, Openai api, Claude Code, Kubernetes, Terraform, Alpaca API, yfinance, Qiskit, PyTorch, TensorFlow, ETL pipelines, CI/CD, HPC/cluster computing, statistical/financial analysis, **Languages:** Arabic (Fluent)